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Student Number

Class



## Gosford High School

### Trial HSC 2025

## Mathematics Extension 2

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#### General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black or blue pen.
- Calculators approved by NESA may be used.
- A reference sheet is provided.
- In Section II, show relevant mathematical reasoning and/ or calculations.

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#### Total Marks

100

#### Section I – 10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section
- Answer questions on the Multiple-choice answer sheet

#### Section II – 90 marks

- Attempt Questions 11–16
- Allow about 2 hours 45 minutes for this section
- Answer questions on booklets provided for each question

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Shade the best response on the multiple-choice answer sheet.

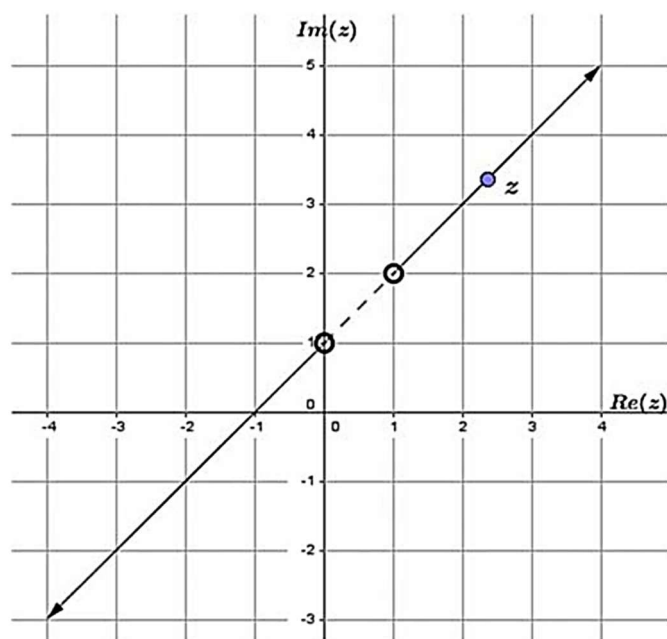
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1. Which of the following statements is a correct interpretation of ‘ $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z}$  such that  $\frac{a}{b} \in \mathbb{Z}$ ’?
- (A) Every integer is also a rational number
  - (B) There exists a rational number that corresponds to each pair of integers
  - (C) Every rational number is also an integer
  - (D) Every integer has an integer factor
2. Which of the following is a unit vector in the opposite direction to  $-\underline{i} + 2\underline{j} - 2\underline{k}$
- (A)  $\frac{1}{\sqrt{5}}(-\underline{i} + 2\underline{j} - 2\underline{k})$
  - (B)  $\frac{1}{\sqrt{5}}(\underline{i} - 2\underline{j} + 2\underline{k})$
  - (C)  $\frac{1}{3}(\underline{i} - 2\underline{j} + 2\underline{k})$
  - (D)  $\frac{1}{3}(-\underline{i} + 2\underline{j} - 2\underline{k})$
3. Which of the following is a primitive of  $x\sin(x)$ ?
- (A)  $\sin(x) - x\cos(x)$
  - (B)  $x\cos(x) - \sin(x)$
  - (C)  $-x\cos(x) - \sin(x)$
  - (D)  $x\cos(x) + \sin(x)$

4. What is the Cartesian equation of the curve defined by  $\mathbf{r}(t) = \begin{pmatrix} \sin t \\ \cos t \\ \cos(2t) \end{pmatrix}$ ?

- (A)  $z = y^2 - x^2$
- (B)  $z = 1 - 2x^2$
- (C)  $x^2 + y^2 = 1$
- (D)  $z = 2y^2 - 1$

5. Which of the following defines the locus of the complex number  $z$  sketched in the diagram below?

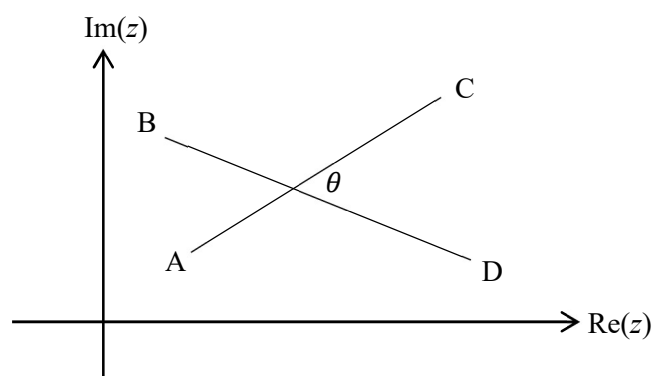


- (A)  $\arg\left(\frac{z - (1 + 2i)}{z - i}\right) = \pi$
- (B)  $\arg\left(\frac{z - i}{z - (1 + 2i)}\right) = \pi$
- (C)  $\arg(z - i) = \arg(z - (1 + 2i))$
- (D)  $\arg(z + i) = \arg(z - (1 + 2i))$

6. Which of the following statements is the negation of ' $\forall x, y \in \mathbb{R}, xy \leq 9 \Rightarrow x \leq 3$  or  $y \leq 3$ '?

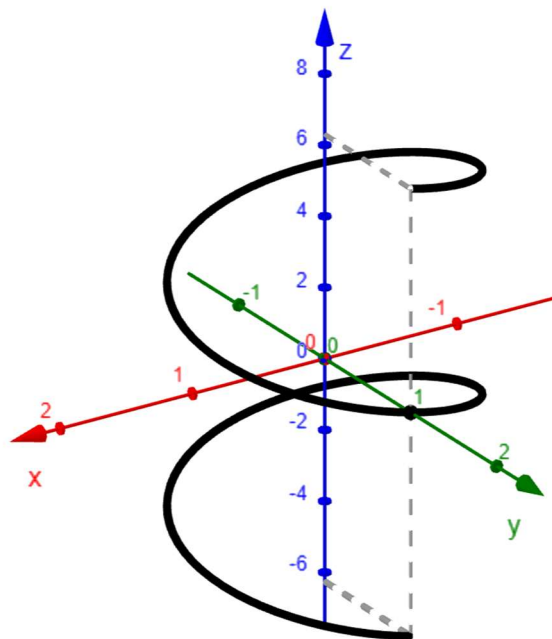
- (A)  $\forall x, y \in \mathbb{R} \ x \leq 3 \text{ or } y \leq 3 \Rightarrow xy \leq 9$
- (B)  $\forall x, y \in \mathbb{R} \ x > 3 \text{ and } y > 3 \Rightarrow xy > 9$
- (C)  $\exists x, y \in \mathbb{R} \text{ with } x > 3 \text{ or } y > 3 \text{ such that } xy \leq 9$
- (D)  $\exists x, y \in \mathbb{R} \text{ with } x > 3 \text{ and } y > 3 \text{ such that } xy \leq 9$

7. The points  $A$ ,  $B$ ,  $C$  and  $D$  on the Argand diagram below represent the complex numbers  $a$ ,  $b$ ,  $c$ , and  $d$ , respectively. The angle  $\theta$  is the acute angle between  $AC$  and  $BD$ . Which of the following is a correct expression for  $\theta$  ?



- (A)  $\theta = \arg\left(\frac{c-d}{a-b}\right)$
- (B)  $\theta = \arg\left(\frac{a-b}{c-d}\right)$
- (C)  $\theta = \arg\left(\frac{c-a}{d-b}\right)$
- (D)  $\theta = \arg\left(\frac{d-b}{c-a}\right)$
8. If  $\frac{dy}{dx} = (1 + \ln x)y$  and when  $x = 1$ ,  $y = 1$  then  $y = ?$
- (A)  $e^{x \ln x}$
- (B)  $1 + \ln x$
- (C)  $e^{2x + x \ln x - 2}$
- (D)  $e^{\frac{x^2 - 1}{x^2}}$

9. Which of the following could be the parametric representation of the curve below?



- (A)  $x = t^2, y = t, z = 1$
- (B)  $x = \sin t, y = \cos(2t), z = t$
- (C)  $x = \sin t, y = \cos t, z = t$
- (D)  $x = \sin t, y = \cos t, z = 2t$
10. Consider the complex number  $z$  such that  $|z - z_1| = \sqrt{5}$  where  $z_1 = \frac{7 + 4i}{3 - 2i}$ . The largest possible value for  $|z|$  is
- (A)  $2\sqrt{5}$
- (B)  $\sqrt{5}$
- (C)  $\sqrt{\frac{11}{5}}$
- (D)  $\sqrt{65}$

Section II begins on the next page.

## Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours 45 minutes for this section.

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### Question 11 (16 marks)

Marks

*Answer in the booklet labelled Question 11*

- (a) Given that  $z = \sqrt{3} - i$ , find:
- (i)  $z - \bar{z}$  1
  - (ii)  $z^{-1}$  1
- (b) Prove or disprove the statement  $3^n + 2$  is prime for all integers  $n \geq 0$  2
- (c) Find  $\int 3x^2 \tan^{-1}(x^3) dx$  3
- (d) Determine the square roots of  $-15 + 8i$ . Show all relevant working. 2
- (e) For two vectors  $\mathbf{m}$  and  $\mathbf{p}$ , it is known that  $|\mathbf{m}| = 5$ ,  $|\mathbf{p}| = 3$ , and  $\mathbf{m} \cdot \mathbf{p} = 9$ . Find  $|\mathbf{m} - \mathbf{p}|$ . 2
- (f) Evaluate  $\int_1^2 \frac{1}{x^2(x+2)} dx$  3
- (g) Prove by contradiction that there are no integers  $a, b$ , such that  $25a + 15b = 1$ . 2

Question 12 begins on the next page

*Answer in the booklet labelled Question 12*

- (a) Represent  $z$  on an Argand diagram, where  $\frac{\pi}{3} \leq \arg(z) \leq \frac{\pi}{2}$  **2**
- (b) (i) Find values of  $A$  and  $B$  such that  $A(\cos x + \sin x) + B(\cos x - \sin x) \equiv 4\sin x$  **1**
- (ii) Hence, or otherwise, find  $\int_0^{\frac{\pi}{4}} \frac{4\sin x}{\cos x + \sin x} dx$  **3**
- (c) If  $\frac{3 + 2i\sin\theta}{1 - 2i\sin\theta}$  is purely imaginary, find  $\theta$  where  $-\pi < \theta \leq \pi$  **2**
- (d) Prove that  $2^{2n} - 1$  is divisible by 3 for all positive integers  $n$ . **2**
- (e) Find the point of intersection of the lines  $\mathbf{r} = \begin{pmatrix} 0 \\ 3 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$  and  $\mathbf{p} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$  **3**
- (f) Consider the sequence defined by the recurrence relation  $t_n = 2t_{n-1} + 3$  for  $n \geq 2$  where  $t_1 = 3$ .  
Prove by mathematical induction that  $t_n = 3 \times 2^n - 3$  for  $n \geq 1$ . **3**

Question 13 begins on the next page

*Answer in the booklet labelled Question 13*

- (a) The point P (2,3) represents the complex number  $2 + 3i$  on the Argand diagram. The line segment OP is rotated anticlockwise about the origin by  $60^\circ$  to the location of point Q. Determine the coordinates of point Q. **2**
- (b) Relative to a fixed origin  $O$ , the points  $A$ ,  $B$  and  $D$  have position vectors  $(2\mathbf{i} - \mathbf{j} + 5\mathbf{k})$ ,  $(5\mathbf{i} + 2\mathbf{j} + 10\mathbf{k})$  and  $(-\mathbf{i} + \mathbf{j} + 4\mathbf{k})$  respectively. The line  $l$  passes through the points  $A$  and  $B$ .
- (i) Find the vector equation of line  $l$ . **1**
- (ii) The points  $A, B, C, D$  form a parallelogram when joined in the given order. Find the position vector of point  $C$ . **2**
- (iii) Determine the area of the parallelogram ABCD, correct to 3 significant figures. **4**
- (c) Evaluate  $\int_0^{\frac{\pi}{3}} \cos^5 x \, dx$  **3**
- (d) If  $a, b > 0$  and  $a + b = 1$  prove that the minimum value of  $\frac{1}{a} + \frac{1}{b}$  is 4. **3**

Question 14 begins on the next page

*Answer in the booklet labelled Question 14*

- (a) (i) Solve  $z^5 + 1 = 0$  **1**
- (ii) Hence express  $z^5 + 1$  as a product of real linear and quadratic factors **2**
- (iii) Hence deduce that  $4\sin\left(\frac{\pi}{10}\right)\cos\left(\frac{\pi}{5}\right) = 1$  **2**
- (b) Find  $\int \frac{x^3}{x^2 + 2x + 5} dx$  **4**
- (c) For positive  $a$  and  $b$ , show by mathematical induction that  $\left(\frac{a+b}{2}\right)^n \leq \frac{a^n + b^n}{2}$  for all positive integers  $n$ . **3**
- (d) Sketch the locus of  $w$  if  $w = \frac{z-1}{z}$ , given that  $|z| = 2$  **4**

Question 15 begins on the next page

*Answer in the booklet labelled Question 15*

- (a) By considering a suitable trigonometric substitution, or otherwise, find  $\int \frac{9}{(9-x^2)^{\frac{3}{2}}} dx$  **4**
- (b) For positive  $x$ ,  $y$ , and  $z$ , prove that  $x+y+z+\frac{1}{4x}+\frac{1}{4y}+\frac{1}{4z} \geq 3$  **3**
- (c) If  $I_n = \int_0^1 (1-x^2)^{\frac{n}{2}} dx$
- (i) Show that  $I_n = \frac{n}{n+1} I_{n-2}$  **3**
- (ii) Hence evaluate  $I_5$  **2**

Question 16 begins on the next page

*Answer in the booklet labelled Question 16*

- (a) (i) Show that  $\sin x > \frac{2x}{\pi}$  for  $0 < x < \frac{\pi}{2}$  2
- (ii) Explain why  $\int_0^{\frac{\pi}{2}} e^{-\sin x} dx < \int_0^{\frac{\pi}{2}} e^{-\frac{2x}{\pi}} dx$  2
- (iii) Show that  $\int_{\frac{\pi}{2}}^{\pi} e^{-\sin x} dx = \int_0^{\frac{\pi}{2}} e^{-\sin x} dx$  2
- (iv) Hence show that  $\int_0^{\pi} e^{-\sin x} dx < \frac{\pi}{e}(e-1)$  2
- (b) In the Argand diagram, the points A, B and C represent the triangle ABC.  
 The vectors  $\overrightarrow{AB}$  and  $\overrightarrow{AC}$  are represented by the complex numbers  $z_1$  and  $z_2$  respectively  
 with  $\text{Arg}(z_1) = \alpha$   $\text{Arg}(z_2) = \beta$  and  $\angle BAC = \theta$ .  
 By expressing  $z_1$  and  $z_2$  in Mod-Arg form, or otherwise, show that the area of triangle  
 ABC is given by  $\text{Area} = \left| \frac{1}{2} \text{Im}(z_1 \overline{z_2}) \right|$  3
- (c) By establishing a reduction formula for  $I_n = \int (\sin^{-1} x)^n dx$ , show that 4
- $$\int_0^1 (\sin^{-1} x)^3 dx = \frac{\pi^3}{8} - 3\pi + 6$$

**End of Exam**

# SOLUTIONS



## Gosford High School

### Trial HSC 2025

## Mathematics Extension 2

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- Working time – 3 hours
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#### Total Marks

**100**

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- Allow about 15 minutes for this section
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- Allow about 2 hours 45 minutes for this section
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## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Shade the best response on the multiple-choice answer sheet.

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1. Which of the following statements is a correct interpretation of ‘ $\forall a \in \mathbb{Z}, \exists b \in \mathbb{Z}$  such that  $\frac{a}{b} \in \mathbb{Z}$ ’?
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- (A)  $\sin(x) - x\cos(x)$
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4. What is the Cartesian equation of the curve defined by  $\mathbf{r}(t) = \begin{pmatrix} \sin t \\ \cos t \\ \cos(2t) \end{pmatrix}$ ?

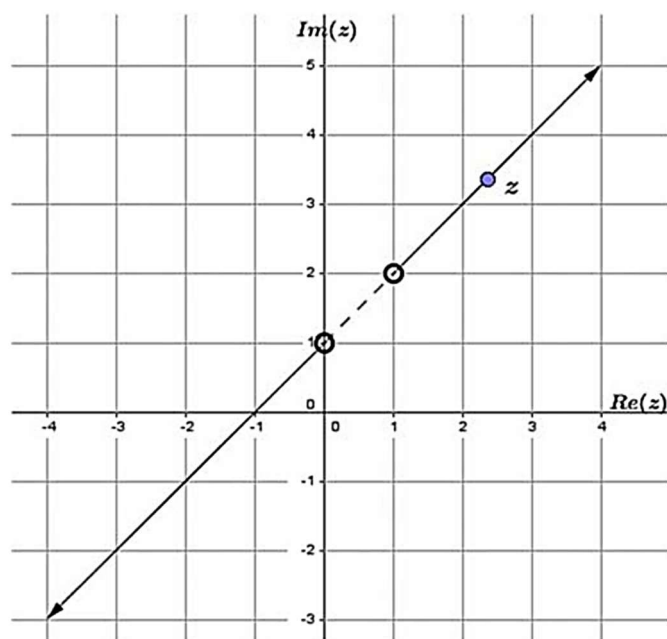
(A)  $z = y^2 - x^2$

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(C)  $x^2 + y^2 = 1$

(D)  $z = 2y^2 - 1$

5. Which of the following defines the locus of the complex number  $z$  sketched in the diagram below?



(A)  $\arg\left(\frac{z - (1 + 2i)}{z - i}\right) = \pi$

(B)  $\arg\left(\frac{z - i}{z - (1 + 2i)}\right) = \pi$

(C)  $\arg(z - i) = \arg(z - (1 + 2i))$

(D)  $\arg(z + i) = \arg(z - (1 + 2i))$

6. Which of the following statements is the negation of ' $\forall x, y \in \mathbb{R}, xy \leq 9 \Rightarrow x \leq 3$  or  $y \leq 3$ '?

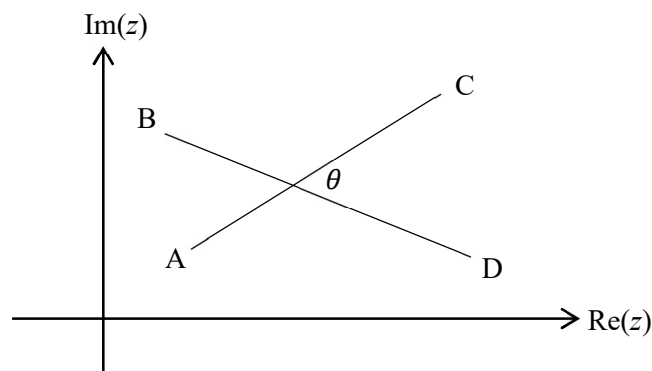
(A)  $\forall x, y \in \mathbb{R} \ x \leq 3$  or  $y \leq 3 \Rightarrow xy \leq 9$

(B)  $\forall x, y \in \mathbb{R} \ x > 3$  and  $y > 3 \Rightarrow xy > 9$

(C)  $\exists x, y \in \mathbb{R}$  with  $x > 3$  or  $y > 3$  such that  $xy \leq 9$

(D)  $\exists x, y \in \mathbb{R}$  with  $x > 3$  and  $y > 3$  such that  $xy \leq 9$

7. The points  $A$ ,  $B$ ,  $C$  and  $D$  on the Argand diagram below represent the complex numbers  $a$ ,  $b$ ,  $c$ , and  $d$ , respectively. The angle  $\theta$  is the acute angle between  $AC$  and  $BD$ . Which of the following is a correct expression for  $\theta$  ?

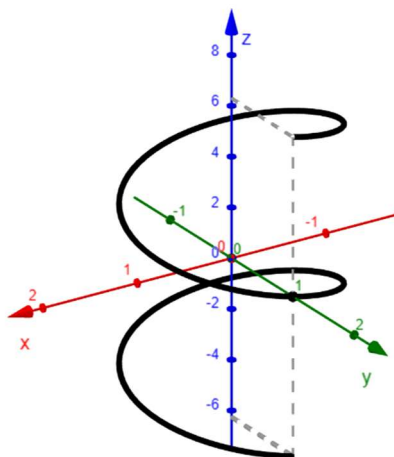


- (A)  $\theta = \arg\left(\frac{c-d}{a-b}\right)$
- (B)  $\theta = \arg\left(\frac{a-b}{c-d}\right)$
- (C)  $\theta = \arg\left(\frac{c-a}{d-b}\right)$
- (D)  $\theta = \arg\left(\frac{d-b}{c-a}\right)$

8. If  $\frac{dy}{dx} = (1 + \ln x)y$  and when  $x = 1$ ,  $y = 1$  then  $y = ?$

- (A)  $e^{x \ln x}$
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- (D)  $e^{\frac{x^2 - 1}{x^2}}$

9. Which of the following could be the parametric representation of the curve below?



- (A)  $x = t^2, y = t, z = 1$
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10. Consider the complex number  $z$  such that  $|z - z_1| = \sqrt{5}$  where  $z_1 = \frac{7 + 4i}{3 - 2i}$ .  
The largest possible value for  $|z|$  is
- (A)  $2\sqrt{5}$
- (B)  $\sqrt{5}$
- (C)  $\sqrt{\frac{11}{5}}$
- (D)  $\sqrt{65}$

Section II begins on the next page.

**Section II****90 marks****Attempt Questions 11-16****Allow about 2 hours 45 minutes for this section.****Question 11 (16 marks)****Marks***Answer in the booklet labelled Question 11*(a) Given that  $z = \sqrt{3} - i$ , find:

(i)  $z - \bar{z}$

**1***Sample solution*

$$z - \bar{z} = 2 \operatorname{Im}(z) \\ = -2$$

| Marks | Criteria         |
|-------|------------------|
| 1     | Correct solution |

(ii)  $z^{-1}$

**1***Sample solution*

$$z^{-1} = \frac{1}{\sqrt{3} - i} \times \frac{\sqrt{3} + i}{\sqrt{3} + i} \\ = \frac{\sqrt{3} + i}{3 + 1} \\ = \frac{1}{4}(\sqrt{3} + i)$$

| Marks | Criteria         |
|-------|------------------|
| 1     | Correct solution |

(b) Prove or disprove the statement  $3^n + 2$  is prime for all integers  $n \geq 0$ **2***Sample solution*

Let  $n = 5$   $3^5 + 2 = 243 + 2$  which is not prime. Hence disproved by counter example.

$$= 245$$

$$= 5 \times 49$$

| Marks | Criteria                                                                                    |
|-------|---------------------------------------------------------------------------------------------|
| 2     | Correct solution, including identifying that the statement is disproven by counter-example  |
| 1     | Counter example found but not identified as a proof by counter-example, or equivalent merit |

(c) Find  $\int 3x^2 \tan^{-1}(x^3) dx$

3

*Sample solution*

$$\begin{aligned} \text{Let } u &= \tan^{-1}(x^3) & v &= x^3 \\ u' &= 3x^2 \times \frac{1}{1+x^6} & v' &= 3x^2 \\ &= \frac{3x^2}{1+x^6} \end{aligned}$$

$$\begin{aligned} \int 3x^2 \tan^{-1}(x^3) dx &= x^3 \tan^{-1}(x^3) - \int \frac{3x^5}{1+x^6} dx \\ &= x^3 \tan^{-1}(x^3) - \frac{1}{2} \ln|1+x^6| + C \\ &= x^3 \tan^{-1}(x^3) - \frac{1}{2} \ln(1+x^6) + C \end{aligned}$$

| Marks | Criteria                                                                                                                                |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 3     | Correct solution                                                                                                                        |
| 2     | Correctly applies integration by parts formula, but an error prevents arrival at correct solution, or equivalent merit.                 |
| 1     | Differentiates one factor and integrates the other factor with the intention of applying differentiation by parts, or equivalent merit. |

(d) Determine the square roots of  $-15 + 8i$ . Show all relevant working.

2

*Sample solution*

Let  $(a + bi)^2 = -15 + 8i$

$$a^2 - b^2 = -15$$

$$2ab = 8$$

$$ab = 4$$

By inspection,  $a = \pm 1$ ,  $b = \pm 4$

Hence  $\sqrt{-15 + 8i} = \pm(1 + 4i)$

| Marks | Criteria                                                       |
|-------|----------------------------------------------------------------|
| 2     | Correct solution                                               |
| 1     | Obtains a pair of simultaneous equations, or equivalent merit. |

- (e) For two vectors  $\underline{m}$  and  $\underline{p}$ , it is known that  $|\underline{m}| = 5$ ,  $|\underline{p}| = 3$ , and  $\underline{m} \cdot \underline{p} = 9$ . Find  $|\underline{m} - \underline{p}|$ .

2

Sample solution

Method 1

$$\begin{aligned} |\underline{m} - \underline{p}|^2 &= (\underline{m} - \underline{p}) \cdot (\underline{m} - \underline{p}) \\ &= \underline{m} \cdot \underline{m} - 2\underline{m} \cdot \underline{p} + \underline{p} \cdot \underline{p} \\ &= |\underline{m}|^2 - 2\underline{m} \cdot \underline{p} + |\underline{p}|^2 \\ &= 5^2 - 2(9) + 3^2 \\ &= 16 \\ \therefore |\underline{m} - \underline{p}| &= 4 \end{aligned}$$

Method 2

$$\begin{aligned} \underline{m} \cdot \underline{p} &= |\underline{m}| |\underline{p}| \cos \theta \\ 9 &= 5 \times 3 \times \cos \theta \\ \cos \theta &= \frac{3}{5} \\ |\underline{m} - \underline{p}|^2 &= |\underline{m}|^2 + |\underline{p}|^2 - 2|\underline{m}| |\underline{p}| \cos \theta \\ |\underline{m} - \underline{p}|^2 &= 5^2 + 3^2 - 2(5)(3) \left( \frac{3}{5} \right) \\ |\underline{m} - \underline{p}|^2 &= 16 \\ |\underline{m} - \underline{p}| &= 4 \end{aligned}$$

| Marks | Criteria                                                                                                                       |
|-------|--------------------------------------------------------------------------------------------------------------------------------|
| 2     | Correct solution                                                                                                               |
| 1     | Applies the relationship $ \underline{v} ^2 = \underline{v} \cdot \underline{v}$ OR finds $\cos \theta$ , or equivalent merit. |

- (f) Evaluate  $\int_1^2 \frac{1}{x^2(x+2)} dx$

3

Sample solution

Consider  $\frac{A}{x^2} + \frac{B}{x} + \frac{C}{x+2} \equiv \frac{1}{x^2(x+2)}$

$$A(x+2) + Bx(x+2) + Cx^2 \equiv 1$$

Let  $x = 0$  :  $2A = 1 \Rightarrow A = \frac{1}{2}$

Let  $x = -2$  :  $4C = 1 \Rightarrow C = \frac{1}{4}$

Let  $x = -1$  :  $\frac{1}{2} - B + \frac{1}{4} = 1 \Rightarrow B = -\frac{1}{4}$

$$\begin{aligned} \int_1^2 \frac{1}{x^2(x+2)} dx &= \int_1^2 \frac{1}{2} x^{-2} - \frac{1}{4} \cdot \frac{1}{x} + \frac{1}{4} \cdot \frac{1}{x+2} dx \\ &= \left[ -\frac{1}{2x} - \frac{1}{4} \ln|x| + \frac{1}{4} \ln|x+2| \right]_1^2 \\ &= \left[ -\frac{1}{2x} + \frac{1}{4} \ln \left| \frac{x+2}{x} \right| \right]_1^2 \\ &= \left[ -\frac{1}{4} + \frac{1}{4} \ln(2) \right] - \left[ -\frac{1}{2} + \frac{1}{4} \ln(3) \right] \\ &= \frac{1}{4} + \frac{1}{4} \ln \left( \frac{2}{3} \right) \end{aligned}$$

| Marks | Criteria                                                |
|-------|---------------------------------------------------------|
| 3     | Correct solution                                        |
| 2     | Obtains correct primitive function, or equivalent merit |
| 1     | Decomposes into partial fractions, or equivalent merit  |

(g) Prove by contradiction that there are no integers  $a, b$ , such that  $25a + 15b = 1$ .

2

*Sample solution*

Negation: suppose  $25a + 15b = 1$  where  $a$  and  $b$  are integers.

$$25a = 1 - 15b$$

If  $a$  is an integer then LHS is a multiple of 5. Hence RHS must also be a multiple of 5.

$$\text{Let } -15b + 1 = 5M \text{ where } M \in \mathbb{Z} \Rightarrow b = \frac{5M - 1}{15}$$

But  $5M - 1$  cannot be divisible by 15 as  $5M - 1$  is not divisible by 5. Which leaves  $b \notin \mathbb{Z}$  contradicting the definition of  $b$  being an integer.

The negation is false by contradiction, hence the original statement must be true.

| Marks | Criteria                                                                                                                   |
|-------|----------------------------------------------------------------------------------------------------------------------------|
| 2     | Correct solution                                                                                                           |
| 1     | Sets up appropriate conditions including both sides being multiples of 5 if $a$ and $b$ are integers, or equivalent merit. |

Question 12 begins on the next page

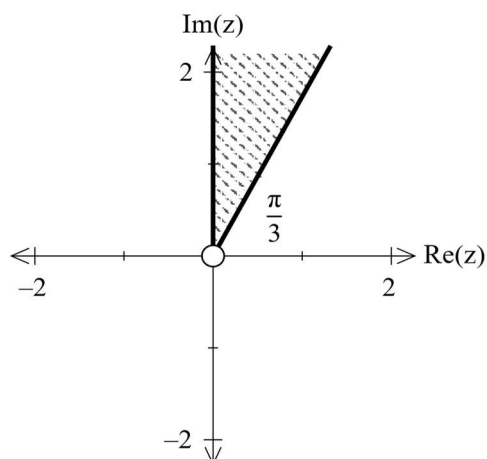
**Question 12 (16 Marks)**

**Marks**

- (a) Represent  $z$  on an Argand diagram, where  $\frac{\pi}{3} \leq \arg(z) \leq \frac{\pi}{2}$

**2**

*Sample solution*



| Marks | Criteria                          |
|-------|-----------------------------------|
| 2     | Correct solution                  |
| 1     | Progress towards correct solution |

- (b) (i) Find values of  $A$  and  $B$  such that  $A(\cos x + \sin x) + B(\cos x - \sin x) \equiv 4\sin x$

**1**

*Sample solution*

$$A(\cos x + \sin x) + B(\cos x - \sin x) \equiv 4\sin x$$

$$(A + B)\cos x + (A - B)\sin x \equiv 4\sin x$$

Equating coefficients on LHS and RHS:

$$A + B = 0$$

$$A - B = 4$$

$$\therefore A = 2, B = -2$$

| Marks | Criteria         |
|-------|------------------|
| 1     | Correct solution |

(ii) Hence, or otherwise, find  $\int_0^{\frac{\pi}{4}} \frac{4\sin x}{\cos x + \sin x} dx$

3

*Sample solution*

$$\begin{aligned} \int_0^{\frac{\pi}{4}} \frac{4\sin x}{\cos x + \sin x} dx &= \int_0^{\frac{\pi}{4}} \frac{2(\cos x + \sin x)}{\cos x + \sin x} - \frac{2(\cos x - \sin x)}{\cos x + \sin x} dx \\ &= \int_0^{\frac{\pi}{4}} 2 - 2 \frac{-\sin x + \cos x}{\cos x + \sin x} dx \\ &= \left[ 2x - 2 \ln |\cos x + \sin x| \right]_0^{\frac{\pi}{4}} \\ &= \left[ 2\left(\frac{\pi}{4}\right) - 2 \ln \left| \cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right) \right| \right] - \left[ 2(0) - 2 \ln |\cos(0) + \sin(0)| \right] \\ &= \frac{\pi}{2} - 2 \ln \left( \frac{2}{\sqrt{2}} \right) - 0 - \ln(1) \\ &= \frac{\pi}{2} - \ln(2) \end{aligned}$$

| Marks | Criteria                           |
|-------|------------------------------------|
| 3     | Correct solution                   |
| 2     | Obtains correct primitive function |
| 1     | Applies result from part (a)       |

(c) If  $\frac{3 + 2i\sin\theta}{1 - 2i\sin\theta}$  is purely imaginary, find  $\theta$  where  $-\pi < \theta \leq \pi$

2

*Sample solution*

$$\frac{3 + 2i\sin\theta}{1 - 2i\sin\theta} \times \frac{1 + 2i\sin\theta}{1 + 2i\sin\theta} = \frac{3 + 8i\sin\theta - 4\sin^2\theta}{1 + 4\sin^2\theta}$$

Purely imaginary  $\Rightarrow$  Real part is zero.

$$3 - 4\sin^2\theta = 0$$

$$\sin\theta = \pm \frac{\sqrt{3}}{2}$$

$$\therefore \theta = \pm \frac{\pi}{3}, \pm \frac{2\pi}{3}$$

| Marks | Criteria                                                                                       |
|-------|------------------------------------------------------------------------------------------------|
| 2     | Correct solution                                                                               |
| 1     | Progress towards correct solution. I.e. realises the denominator and simplifies the numerator. |

(d) Prove that  $2^{2n} - 1$  is divisible by 3 for all positive integers  $n$ .

2

*Sample solution*

$2^{2n} - 1 = (2^n - 1)(2^n + 1)$  by difference of two squares.

Now  $2^n - 1$ ,  $2^n$  and  $2^n + 1$  are three consecutive integers, hence one of them must be divisible by 3.

$2^n$  is not divisible by 3. Hence either  $2^n - 1$  or  $2^n + 1$  must be divisible by 3.

$\therefore (2^n - 1)(2^n + 1)$  is divisible by 3 and hence  $2^{2n} - 1$  is divisible by 3.

| Marks | Criteria                          |
|-------|-----------------------------------|
| 2     | Correct solution                  |
| 1     | Progress towards correct solution |

(e) Find the point of intersection of the lines  $\mathbf{r} = \begin{pmatrix} 0 \\ 3 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$  and  $\mathbf{p} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$

3

*Sample solution*

$$x \text{ component: } 0 - \lambda = -1 - \mu \quad \therefore \mu - \lambda = -1 \quad \textcircled{1}$$

$$y \text{ component: } 3 + \lambda = 2 + 2\mu \quad \therefore 2\mu - \lambda = 1 \quad \textcircled{2}$$

$$z \text{ component: } -5 + 2\lambda = 3 - \mu \quad \therefore \mu + 2\lambda = 8 \quad \textcircled{3}$$

$$\textcircled{2} - \textcircled{1} : \quad \mu = 2$$

$$\therefore \quad \lambda = 3$$

Check for consistency in  $\textcircled{3} : (2) + 2(3) = 8 = RHS$

$\therefore$  point of intersection occurs where  $\lambda = 3$  and  $\mu = 2$

$$\text{Point of intersection is } \begin{pmatrix} -3 \\ 6 \\ 1 \end{pmatrix}$$

| Marks | Criteria                                                                                                                                        |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | Correct solution                                                                                                                                |
| 2     | Finds the values of $\mu$ and $\lambda$ , or makes an error in solving the simultaneous equations but goes onto find the point of intersection. |
| 1     | Obtains a pair of simultaneous equations that may lead to finding $\mu$ or $\lambda$                                                            |

(f) Consider the sequence defined by the recurrence relation  $t_n = 2t_{n-1} + 3$  for  $n \geq 2$  where  $t_1 = 3$ .

Prove by mathematical induction that  $t_n = 3 \times 2^n - 3$  for  $n \geq 1$ .

**3**

*Sample solution*

For  $n = 1$  :  $t_1 = 3 \times 2^1 - 3$

$= 3$  which is the given value for  $t_1$ .  $\therefore$  true for  $n = 1$

Assume true for  $n = k$  :  $t_k = 3 \times 2^k - 3$  where  $k$  is an integer.

For  $n = k + 1$ , *R.T.P.*  $t_{k+1} = 3 \times 2^{k+1} - 3$

$LHS = t_{k+1}$

$= 2t_k + 3$  by recurrence definition

$= 2(3 \times 2^k - 3) + 3$  by assumption

$= 3 \times 2^{k+1} - 6 + 3$

$= 3 \times 2^{k+1} - 3$

$= RHS$  as required.

$\therefore$  true for integers  $n \geq 1$  by process of mathematical induction.

| Marks            | Criteria                                                                              |
|------------------|---------------------------------------------------------------------------------------|
| 3                | Correct solution                                                                      |
| Allow 1 for each | Incorporates the assumption in the induction step                                     |
|                  | Verifies the initial case and applies the recurrence definition in the induction step |

Question 13 begins on the next page

**Question 13 (15 Marks)**

**Marks**

- (a) The point P (2,3) represents the complex number  $2 + 3i$  on the Argand diagram.  
The line segment OP is rotated anticlockwise about the origin by  $60^\circ$  to the location of point Q.  
Determine the coordinates of point Q.

**2**

*Sample solution*

Rotating by  $\frac{\pi}{3}$  anticlockwise about the origin is equivalent to multiplying by  $\text{cis}\left(\frac{\pi}{3}\right)$ .

$$\begin{aligned}(2 + 3i)\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right) &= (2 + 3i)\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) \\ &= \left(1 - \frac{3\sqrt{3}}{2}\right) + \left(\frac{3}{2} + \sqrt{3}i\right) \\ \therefore Q \text{ has coordinates } &\left(\left(1 - \frac{3\sqrt{3}}{2}\right), \left(\frac{3}{2} + \sqrt{3}\right)\right)\end{aligned}$$

| Marks | Criteria                                                                                                                    |
|-------|-----------------------------------------------------------------------------------------------------------------------------|
| 2     | Correct solution                                                                                                            |
| 1     | Recognises that multiplying by $\text{cis}\left(\frac{\pi}{3}\right)$ generates the required rotation, or equivalent merit. |

- (b) Relative to a fixed origin  $O$ , the points  $A$ ,  $B$  and  $D$  have position vectors  $(2\mathbf{i} - \mathbf{j} + 5\mathbf{k})$ ,  $(5\mathbf{i} + 2\mathbf{j} + 10\mathbf{k})$  and  $(-\mathbf{i} + \mathbf{j} + 4\mathbf{k})$  respectively. The line  $l$  passes through the points  $A$  and  $B$ .

- (i) Find the vector equation of line  $l$ .

**1**

*Sample solution*

$$\begin{aligned}l &= \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 5-2 \\ 2+1 \\ 10-5 \end{pmatrix} \\ \therefore l &= \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 3 \\ 5 \end{pmatrix}\end{aligned}$$

| Marks | Criteria         |
|-------|------------------|
| 1     | Correct solution |

(ii) The points  $A, B, C, D$  form a parallelogram when joined in the given order.

Find the position vector of point  $C$ .

2

Sample solution

$$\begin{aligned} \overrightarrow{DC} &= \overrightarrow{AB} \\ \overrightarrow{OC} - \overrightarrow{OD} &= \overrightarrow{OB} - \overrightarrow{OA} \\ \text{Let } C \text{ be } \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \begin{pmatrix} x+1 \\ y-1 \\ z-4 \end{pmatrix} &= \begin{pmatrix} 5-2 \\ 2+1 \\ 10-5 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ 3 \\ 5 \end{pmatrix} \\ \therefore C &= \begin{pmatrix} 2 \\ 4 \\ 9 \end{pmatrix} \end{aligned}$$

| Marks | Criteria                              |
|-------|---------------------------------------|
| 2     | Correct solution                      |
| 1     | Attempts to equate equivalent vectors |

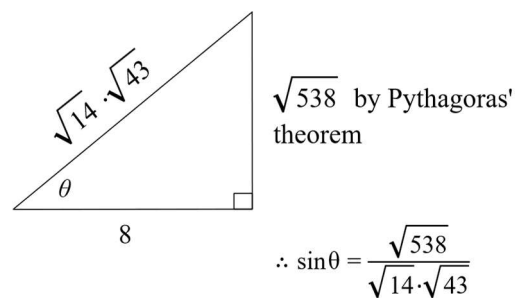
(iii) Determine the area of the parallelogram  $ABCD$ , correct to 3 significant figures.

4

Sample solution

Let  $\theta$  be the angle between the tails of  $\overrightarrow{DA}$  and  $\overrightarrow{DC}$

$$\begin{aligned} \cos \theta &= \frac{\overrightarrow{DA} \cdot \overrightarrow{DC}}{|\overrightarrow{DA}| |\overrightarrow{DC}|} \\ &= \frac{\begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 3 \\ 5 \end{pmatrix}}{\sqrt{14} \cdot \sqrt{43}} \\ &= \frac{8}{\sqrt{14} \cdot \sqrt{43}} \\ \text{Area} &= 2 \times \frac{1}{2} |\overrightarrow{DA}| |\overrightarrow{DC}| \sin \theta \\ &= \sqrt{14} \sqrt{43} \frac{\sqrt{538}}{\sqrt{14} \sqrt{43}} \\ &= \sqrt{538} \text{ sq units} \\ &(\approx 23.19 \text{ sq units}) \end{aligned}$$



| Marks | Criteria                                                      |
|-------|---------------------------------------------------------------|
| 4     | Correct solution                                              |
| 3     | Significant progress towards solution                         |
| 2     | Finds the angle, or the cosine of the angle between two sides |
| 1     | Finds the length of side of the parallelogram                 |

(c) Evaluate  $\int_0^{\frac{\pi}{3}} \cos^5 x \, dx$

3

*Sample solution*

$$\begin{aligned}
 \int_0^{\frac{\pi}{3}} \cos^5 x \, dx &= \int_0^{\frac{\pi}{3}} \cos x (\cos^4 x) \, dx \\
 &= \int_0^{\frac{\pi}{3}} \cos x (1 - \sin^2 x)^2 \, dx \\
 &= \int_0^{\frac{\pi}{3}} \cos x (1 - 2\sin^2 x + \sin^4 x) \, dx \\
 &= \int_0^{\frac{\pi}{3}} (\cos x - 2\cos x \sin^2 x + \cos x \sin^4 x) \, dx \\
 &= \left[ \sin x - \frac{2\sin^3 x}{3} + \frac{\sin^5 x}{5} \right]_0^{\frac{\pi}{3}} \\
 &= \left[ \sin\left(\frac{\pi}{3}\right) - \frac{2}{3} \left( \sin\left(\frac{\pi}{3}\right) \right)^3 + \frac{1}{5} \left( \sin\left(\frac{\pi}{3}\right) \right)^5 \right] - \left[ \sin(0) - \frac{2}{3} (\sin(0))^2 + \frac{1}{5} (\sin(0))^5 \right] \\
 &= \frac{\sqrt{3}}{2} - \frac{2}{3} \times \frac{3\sqrt{3}}{8} + \frac{1}{5} \times \frac{9\sqrt{3}}{32} \\
 &= \frac{49\sqrt{3}}{160}
 \end{aligned}$$

| Marks | Criteria                                                |
|-------|---------------------------------------------------------|
| 3     | Correct solution                                        |
| 2     | Finds correct primitive function                        |
| 1     | Expresses the integral in a form that can be integrated |

(d) If  $a, b > 0$  and  $a + b = 1$  prove that the minimum value of  $\frac{1}{a} + \frac{1}{b}$  is 4.

3

Sample solution

$$\frac{\frac{1}{a} + \frac{1}{b}}{2} \geq \sqrt{\frac{1}{ab}} \quad (\text{AM/GM inequality})$$

$$\therefore \frac{1}{a} + \frac{1}{b} \geq \frac{2}{\sqrt{ab}}$$

Now,  $\frac{2}{\sqrt{ab}}$  is minimised when  $ab$  is maximised for positive  $a, b$ .

Furthermore,  $ab = a(1 - a)$  since  $a + b = 1$ . The maximum value of  $ab$  occurs when  $a = \frac{1}{2}$  by the symmetry of the quadratic function  $a(1 - a)$ .

$$\begin{aligned} \text{Max}[ab] &= \frac{1}{2} \left( 1 - \frac{1}{2} \right) \\ &= \frac{1}{4} \end{aligned}$$

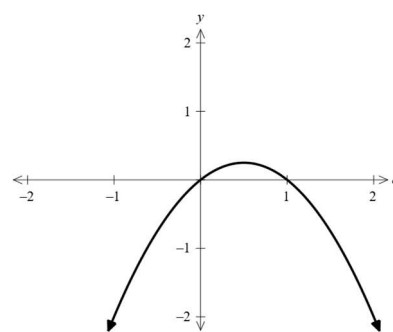
$$\text{Hence } \text{Max}[\sqrt{ab}] = \frac{1}{2}$$

$$\begin{aligned} \text{Hence } \text{Min}\left[\frac{2}{\sqrt{ab}}\right] &= \frac{2}{\frac{1}{2}} \\ &= 4 \end{aligned}$$

$$\therefore \frac{1}{a} + \frac{1}{b} \geq 4$$

$$\text{i.e. } \frac{1}{a} + \frac{1}{b} \geq \frac{2}{\sqrt{ab}} \geq 4$$

$$\therefore \frac{1}{a} + \frac{1}{b} \geq 4 \text{ as required.}$$



| Marks             | Criteria                                                                                   |
|-------------------|--------------------------------------------------------------------------------------------|
| 3                 | Correct solution                                                                           |
| Allow 1 each for: | Applies relationship $\frac{1}{x}$ is minimised when $x$ is maximised, or equivalent merit |
|                   | Applies AM/GM inequality, or equivalent merit                                              |

Question 14 begins on the next page

**Question 14 (16 Marks)**

**Marks**

(a) (i) Solve  $z^5 + 1 = 0$

**1**

*Sample solution*

$$\begin{aligned} z^5 &= -1 \\ z &= -1, \operatorname{cis}\left(\pi \pm \frac{2\pi}{5}\right), \operatorname{cis}\left(\pi \pm \frac{4\pi}{5}\right) \\ &= -1, \operatorname{cis}\left(\pm \frac{\pi}{5}\right), \operatorname{cis}\left(\pm \frac{3\pi}{5}\right) \end{aligned}$$

| Marks | Criteria         |
|-------|------------------|
| 1     | Correct solution |

(ii) Hence express  $z^5 + 1$  as a product of real linear and quadratic factors

**2**

*Sample solution*

$$\begin{aligned} z^5 + 1 &= (z - (-1)) \left( z - \operatorname{cis}\frac{\pi}{5} \right) \left( z - \operatorname{cis}\left(-\frac{\pi}{5}\right) \right) \left( z - \operatorname{cis}\frac{3\pi}{5} \right) \left( z - \operatorname{cis}\left(-\frac{3\pi}{5}\right) \right) \\ &= (z + 1) \left( z^2 - \left( 2\cos\frac{\pi}{5} \right) z + 1 \right) \left( z^2 - \left( 2\cos\frac{3\pi}{5} \right) z + 1 \right) \end{aligned}$$

| Marks | Criteria                                           |
|-------|----------------------------------------------------|
| 2     | Correct solution                                   |
| 1     | Expresses $z^5 + 1$ as a product of linear factors |

(iii) Hence deduce that  $4\sin\left(\frac{\pi}{10}\right)\cos\left(\frac{\pi}{5}\right) = 1$

**2**

*Sample solution 1*

Product of the roots of  $z^5 + 1 = 0$  is  $\frac{f}{a} = \frac{1}{1}$

$$\begin{aligned} -\operatorname{cis}\left(\frac{\pi}{5}\right) \operatorname{cis}\left(-\frac{\pi}{5}\right) \operatorname{cis}\left(\frac{3\pi}{5}\right) \operatorname{cis}\left(-\frac{3\pi}{5}\right) &= 1 \\ -2\cos\left(\frac{\pi}{5}\right) \times 2\cos\left(\frac{3\pi}{5}\right) &= 1 \\ -4\cos\left(\frac{\pi}{5}\right) \times \sin\left(\frac{\pi}{2} - \frac{3\pi}{5}\right) &= 1 \\ -4\cos\left(\frac{\pi}{5}\right) \times \sin\left(-\frac{\pi}{10}\right) &= 1 \\ \therefore 4\cos\left(\frac{\pi}{5}\right) \sin\left(\frac{\pi}{10}\right) &= 1 \text{ since } \sin(-\theta) = -\sin\theta \text{ (odd function)} \end{aligned}$$

*Sample solution 2*

$$z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1) \text{ and}$$

$$z^5 + 1 = (z + 1) \left( z^2 - 2\cos\frac{\pi}{5}z + 1 \right) \left( z^2 - 2\cos\frac{3\pi}{5}z + 1 \right) \text{ from (i)}$$

$$\text{Hence } z^4 - z^3 + z^2 - z + 1 = \left( z^2 - 2\cos\frac{\pi}{5}z + 1 \right) \left( z^2 - 2\cos\frac{3\pi}{5}z + 1 \right)$$

Equating the coefficients of  $z^2$  on each side gives

$$1 = 4\cos\frac{\pi}{5}\cos\frac{3\pi}{5} + 1 + 1$$

$$-1 = 4\cos\frac{\pi}{5}\cos\frac{3\pi}{5} \quad \text{①}$$

$$\text{Now } \cos\frac{3\pi}{5} = \sin\left(\frac{\pi}{2} - \frac{3\pi}{5}\right) \text{ by complementary angle result}$$

$$= \sin\left(-\frac{\pi}{10}\right)$$

$$= -\sin\left(\frac{\pi}{10}\right)$$

And substituting this into ① gives:

$$-1 = 4\cos\left(\frac{\pi}{5}\right) \times -\sin\left(\frac{\pi}{10}\right)$$

$$\therefore 4\sin\left(\frac{\pi}{10}\right)\cos\left(\frac{\pi}{5}\right) = 1 \text{ as required.}$$

| Marks | Criteria                              |
|-------|---------------------------------------|
| 2     | Correct solution                      |
| 1     | Significant progress towards solution |

(b) Find  $\int \frac{x^3}{x^2 + 2x + 5} dx$

4

Sample solution

$$\begin{aligned} \int \frac{x^3}{x^2 + 2x + 5} dx &= \int \frac{x(x^2 + 2x + 5)}{x^2 + 2x + 5} + \frac{-2(x^2 + 2x + 5)}{x^2 + 2x + 5} + \frac{-x + 10}{x^2 + 2x + 5} dx \\ &= \int x - 2 - \left(\frac{1}{2}\right) \frac{2x + 2}{x^2 + 2x + 5} + \frac{11}{x^2 + 2x + 5} dx \\ &= \int x - 2 - \left(\frac{1}{2}\right) \frac{2x + 2}{x^2 + 2x + 5} + 11 \frac{1}{2^2 + (x + 1)^2} dx \\ &= \frac{x^2}{2} - 2x - \frac{1}{2} \ln|x^2 + 2x + 5| + \frac{11}{2} \arctan\left(\frac{x + 1}{2}\right) + C \\ &= \left[ \frac{1}{2} \left( x^2 - 4x - \ln|x^2 + 2x + 5| + 11 \arctan\left(\frac{x + 1}{2}\right) \right) \right] + C \end{aligned}$$

| Marks | Criteria                                                                                           |
|-------|----------------------------------------------------------------------------------------------------|
| 4     | Correct solution                                                                                   |
| 3     | Finds the primitive including the $\ln  f(x) $ or $\frac{1}{a} \arctan\left(\frac{f(x)}{a}\right)$ |
| 2     | Expresses the integral in the form required to perform integration                                 |
| 1     | Makes progress with expressing the integral in the required form to allow for integration          |

(c) For positive  $a$  and  $b$ , show by mathematical induction that  $\left(\frac{a+b}{2}\right)^n \leq \frac{a^n + b^n}{2}$  for all positive integers  $n$ .

3

Sample solution

For  $n = 1$  :

$$\begin{aligned} LHS &= \left(\frac{a+b}{2}\right)^1 \\ &= \frac{a+b}{2} \\ RHS &= \frac{a^1 + b^1}{2} \\ &= LHS \quad \therefore \text{true for } n = 1 \end{aligned}$$

Assume true for  $n = k$  :  $\left(\frac{a+b}{2}\right)^k \leq \frac{a^k + b^k}{2}$

For  $n = k + 1$ , RTP  $\left(\frac{a+b}{2}\right)^{k+1} \leq \frac{a^{k+1} + b^{k+1}}{2}$

I.e. RTP  $\left(\frac{a+b}{2}\right)^{k+1} - \frac{a^{k+1} + b^{k+1}}{2} \leq 0$

$$\begin{aligned}
LHS &= \left(\frac{a+b}{2}\right)^{k+1} - \frac{a^{k+1} + b^{k+1}}{2} \\
&= \left(\frac{a+b}{2}\right) \left(\frac{a+b}{2}\right)^k - \frac{a^{k+1} + b^{k+1}}{2} \\
&\leq \left(\frac{a+b}{2}\right) \left(\frac{a^k + b^k}{2}\right) - \frac{a^{k+1} + b^{k+1}}{2} \quad \text{by assumption} \\
&= \frac{a^{k+1} + b^{k+1} + ba^k + ab^k}{4} - \frac{2a^{k+1} + 2b^{k+1}}{4} \\
&= \frac{-a^{k+1} - b^{k+1} + ba^k + ab^k}{4} \\
&= \frac{a^k(b-a) - b^k(b-a)}{4} \\
&= \frac{(b-a)(a^k - b^k)}{4}
\end{aligned}$$

$\leq 0$  for all positive  $a$  and  $b$ .

Case 1 :  $a = b \Rightarrow ((b-a)(a^k - b^k)) = 0$ ,

Case 2 :  $a \neq b \Rightarrow (b-a)$  and  $(a^k - b^k)$  will have opposite signs and the product will be negative.

$\therefore$  true for positive integer values of  $n$  by process of mathematical induction.

| Marks | Criteria                                                                                                                                               |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | Correct solution                                                                                                                                       |
| 2     | Substantial progress towards solution. Accept correct incorporation of assumption then expansion of appropriate brackets and collection of like terms. |
| 1     | Verifies initial case and incorporates the assumption                                                                                                  |

(d) Sketch the locus of  $w$  if  $w = \frac{z-1}{z}$ , given that  $|z| = 2$

4

*Sample solution 1*

$$zw = z - 1$$

$$z(w-1) = -1$$

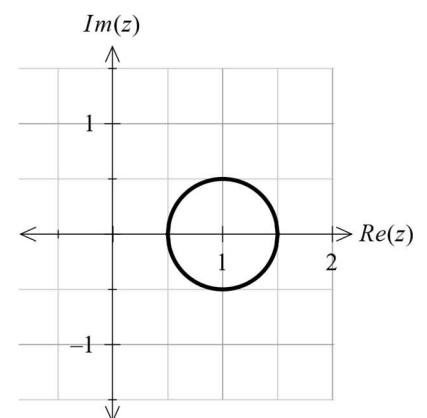
$$z = \frac{1}{1-w}$$

$$|z| = \left| \frac{1}{1-w} \right|$$

$$2 = \frac{1}{|1-w|}$$

$$\therefore |1-w| = \frac{1}{2}$$

$$|w-1| = \frac{1}{2} \text{ which represents a circle centred at } 1 + 0i \text{ with radius } \frac{1}{2}.$$



*Sample solution 2*

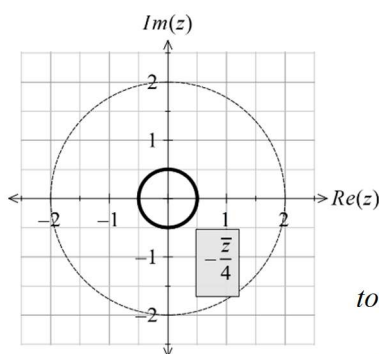
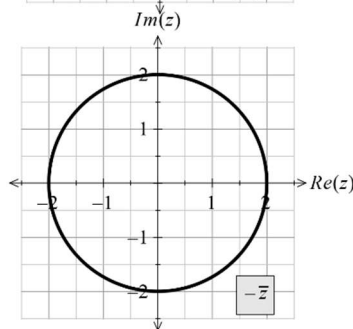
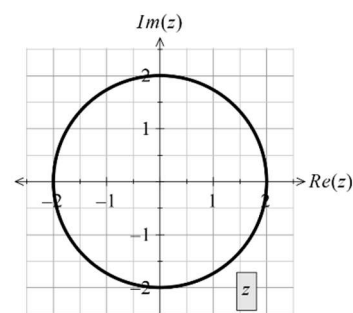
$$\begin{aligned}
 w &= \frac{z-1}{z} \\
 &= 1 - \frac{1}{z} \\
 &= 1 - \frac{\bar{z}}{|z|^2} \\
 &= -\frac{\bar{z}}{4} + 1
 \end{aligned}$$

Locus of  $z$  is a circle centred at the origin with radius 2

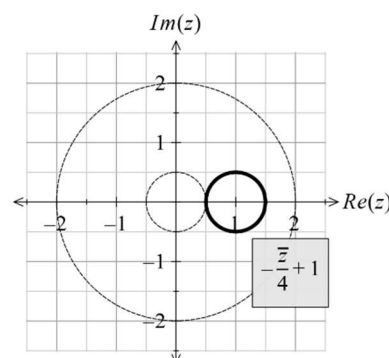
Locus of  $\bar{z}$  is the above circle reflected through the Real axis, i.e. the same circle, and the locus of  $-\left(\frac{\bar{z}}{4}\right)$  will be this same circle rotated by  $\pi$  radians about the origin which still generates the same circle.

Locus of  $-\frac{\bar{z}}{4}$  will have radius  $\frac{1}{4} \times 2 = \frac{1}{2}$

Locus of  $-\frac{\bar{z}}{4} + 1$  will now have 1 added the real part of  $-\frac{\bar{z}}{4}$  thus giving a circle radius  $\frac{1}{2}$  centred at  $1 + 0i$



to  
with



| Marks | Criteria                                                                                                     |
|-------|--------------------------------------------------------------------------------------------------------------|
| 4     | Correct solution, clearly indicating centre and radius of the circle, supported by appropriate calculations. |
| 3     | Significant progress towards solution                                                                        |
| 2     | Expresses $z$ in terms of $w$ , or equivalent merit and incorporates $ z =2$ in a meaningful way             |
| 1     | Expresses $z$ in terms of $w$ , or equivalent merit, or incorporates $ z =2$ in a meaningful way             |

Question 15 begins on the next page

**Question 15 (12 Marks)**

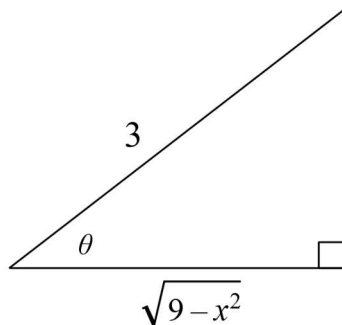
**Marks**

- (a) By considering a suitable trigonometric substitution, or otherwise, find  $\int \frac{9}{(9-x^2)^{\frac{3}{2}}} dx$

**4**

*Sample solution*

$$\begin{aligned}
 \text{Let } x &= 3\sin\theta \\
 \frac{dx}{d\theta} &= 3\cos\theta \\
 dx &= 3\cos\theta \, d\theta
 \end{aligned}
 \qquad
 \begin{aligned}
 \int \frac{9}{(9-x^2)^{\frac{3}{2}}} dx &= \int \frac{9 \times 3\cos\theta}{(9-9\sin^2\theta)^{\frac{3}{2}}} d\theta \\
 &= \int \frac{27\cos\theta}{(9\cos^2\theta)^{\frac{3}{2}}} d\theta \\
 &= \int \frac{27\cos\theta}{27\cos^3\theta} d\theta \\
 &= \int \sec^2\theta \, d\theta \\
 &= \tan\theta + C
 \end{aligned}$$



If  $x = 3\sin\theta$ ,  $\sin\theta = \frac{x}{3}$  and by

Pythagoras theorem,

$$\tan\theta = \frac{x}{\sqrt{9-x^2}}$$

$$\therefore \int \frac{9}{(9-x^2)^{\frac{3}{2}}} dx = \frac{x}{\sqrt{9-x^2}} + C$$

| Marks | Criteria                                                               |
|-------|------------------------------------------------------------------------|
| 4     | Correct solution                                                       |
| 3     | Completes integration in terms of $\theta$ , or equivalent merit       |
| 2     | Partially simplifies the integral but makes no further valid progress. |
| 1     | Completes a valid substitution with change of variable                 |

(b) For positive  $x$ ,  $y$ , and  $z$ , prove that  $x + y + z + \frac{1}{4x} + \frac{1}{4y} + \frac{1}{4z} \geq 3$

3

*Sample solution*

$$\left(\sqrt{a} - \frac{1}{2\sqrt{a}}\right)^2 \geq 0 \text{ for all real } a > 0$$

$$a - 1 + \frac{1}{4a} \geq 0$$

$$\therefore a + \frac{1}{4a} \geq 1$$

This must also be true for  $x$ ,  $y$  and  $z$ , since  $x$ ,  $y$ , and  $z$  are all positive.

$$\therefore x + \frac{1}{4x} \geq 1 \quad \textcircled{1}$$

$$\therefore y + \frac{1}{4y} \geq 1 \quad \textcircled{2}$$

$$\therefore z + \frac{1}{4z} \geq 1 \quad \textcircled{3}$$

① + ② + ③ gives:

$$x + \frac{1}{4x} + y + \frac{1}{4y} + z + \frac{1}{4z} \geq 1 + 1 + 1$$

$$\therefore x + y + z + \frac{1}{4x} + \frac{1}{4y} + \frac{1}{4z} \geq 3 \text{ for positive } x, y, z.$$

| Marks | Criteria                                                         |
|-------|------------------------------------------------------------------|
| 3     | Correct solution                                                 |
| 2     | Substantial progress towards correct solution                    |
| 1     | Notes perfect square is always non-negative, or equivalent merit |

(c) If  $I_n = \int_0^1 (1-x^2)^{\frac{n}{2}} dx$

(i) Show that  $I_n = \frac{n}{n+1} I_{n-2}$

3

*Sample solution*

Let  $u = (1-x^2)^{\frac{n}{2}}$   $v = x$

$u' = -nx(1-x^2)^{\frac{n-2}{2}}$   $v' = 1$

$$I_n = \left[ x(1-x^2)^{\frac{n}{2}} \right]_0^1 - \int_0^1 -nx^2(1-x^2)^{\frac{n-2}{2}} dx$$

$$= 0 + n \int_0^1 (1 - (1-x^2))(1-x^2)^{\frac{n-2}{2}} dx$$

$$= n \int_0^1 (1-x^2)^{\frac{n-2}{2}} - (1-x^2)^{\frac{n}{2}} dx$$

$$= n(I_{n-2} - I_n)$$

$$I_n(1+n) = nI_{n-2}$$

$\therefore I_n = \frac{n}{n+1} I_{n-2}$  as required

| Marks | Criteria                              |
|-------|---------------------------------------|
| 3     | Correct solution                      |
| 2     | Significant progress towards solution |
| 1     | Correct substitution into IBP formula |

(ii) Hence evaluate  $I_5$

2

*Sample solution*

$$I_5 = \frac{5}{6} I_3$$

$$= \frac{5}{6} \times \frac{3}{4} I_1$$

$$= \frac{5}{6} \times \frac{3}{4} \times \int_0^1 (1-x^2)^{\frac{1}{2}} dx$$

$$= \frac{5}{6} \times \frac{3}{4} \times \frac{1}{4} \pi(1)^2 \quad \text{since } \sqrt{1-x^2} \text{ from } x=0 \text{ to } x=1 \text{ forms a quadrant with radius 1.}$$

$$= \frac{5\pi}{32}$$

| Marks | Criteria                                                                   |
|-------|----------------------------------------------------------------------------|
| 2     | Correct solution                                                           |
| 1     | Evaluates $I_1$ or expresses $I_5$ in terms of $I_1$ , or equivalent merit |

Question 16 begins on the next page

**Question 16 (15 Marks)**

**Marks**

- (a) (i) Show that  $\sin x > \frac{2x}{\pi}$  for  $0 < x < \frac{\pi}{2}$

**2**

*Sample solution*

Method 1

Consider the graphs of  $y = \sin x$  and  $y = \frac{2}{\pi}x$  for  $0 < x < \frac{\pi}{2}$ . The graphs have points of intersection

at  $x = 0$  and  $x = \frac{\pi}{2}$  (bounds) and since  $\sin x$  is concave down and  $\frac{2}{\pi}x$  is linear over this domain,

there are no further points of intersection. Hence  $\sin x$  must always be above or below  $\frac{2x}{\pi}$  over this domain.

Consider  $x = \frac{\pi}{4}$  :

$$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}, \quad \frac{2 \times \frac{\pi}{4}}{\pi} = \frac{1}{2} \text{ which is less than } \frac{1}{\sqrt{2}}. \text{ Hence } \sin x > \frac{2x}{\pi} \text{ for } 0 < x < \frac{\pi}{2}$$

| Marks | Criteria                                                |
|-------|---------------------------------------------------------|
| 2     | Correct solution supported by appropriate justification |
| 1     | Progress towards solution                               |

- (ii) Explain why  $\int_0^{\frac{\pi}{2}} e^{-\sin x} dx < \int_0^{\frac{\pi}{2}} e^{-\frac{2x}{\pi}} dx$

**2**

*Sample solution*

$$\sin x > \frac{2x}{\pi} \text{ from part (i)}$$

$$\therefore e^{\sin x} > e^{\frac{2x}{\pi}}$$

$$\frac{1}{e^{\sin x}} < \frac{1}{e^{\frac{2x}{\pi}}} \text{ since both } e^{\sin x} \text{ and } e^{\frac{2x}{\pi}} \text{ are positive for all real } x.$$

$$\therefore e^{-\sin x} < e^{-\frac{2x}{\pi}}$$

$$\text{Hence } \int_0^{\frac{\pi}{2}} e^{-\sin x} dx < \int_0^{\frac{\pi}{2}} e^{-\frac{2x}{\pi}} dx \text{ as required.}$$

| Marks | Criteria                                                |
|-------|---------------------------------------------------------|
| 2     | Correct solution supported by appropriate justification |
| 1     | Progress towards solution                               |

(iii) Show that  $\int_{\frac{\pi}{2}}^{\pi} e^{-\sin x} dx = \int_0^{\frac{\pi}{2}} e^{-\sin x} dx$

2

Sample solution

$$\sin x = \sin(\pi - x)$$

$$\therefore \int_0^{\frac{\pi}{2}} e^{-\sin x} dx = \int_0^{\frac{\pi}{2}} e^{-\sin(\pi-x)} dx \quad (1)$$

Now, let  $u = \pi - x$

Bounds

$$\frac{du}{dx} = -1$$

$$x = \frac{\pi}{2} \rightarrow u = \frac{\pi}{2}$$

$$dx = -1 du$$

$$x = 0 \rightarrow u = \pi$$

Hence

$$\begin{aligned} \int_0^{\frac{\pi}{2}} e^{-\sin(\pi-x)} dx &= \int_{\pi}^{\frac{\pi}{2}} -e^{-\sin u} du \\ &= \int_{\frac{\pi}{2}}^{\pi} e^{-\sin u} du \\ &= \int_{\frac{\pi}{2}}^{\pi} e^{-\sin x} dx \text{ since the variable used is inconsequential.} \quad (2) \end{aligned}$$

Hence from (1) and (2),  $\int_0^{\frac{\pi}{2}} e^{-\sin x} dx = \int_{\frac{\pi}{2}}^{\pi} e^{-\sin x} dx$  as required.

| Marks | Criteria                                             |
|-------|------------------------------------------------------|
| 2     | Correct solution with all links clearly communicated |
| 1     | Progress towards solution                            |

(iv) Hence show that  $\int_0^{\pi} e^{-\sin x} dx < \frac{\pi}{e}(e-1)$

2

*Sample solution*

$$\begin{aligned}
 \int_0^{\pi} e^{-\sin x} dx &= \int_0^{\frac{\pi}{2}} e^{-\sin x} dx + \int_{\frac{\pi}{2}}^{\pi} e^{-\sin x} dx \\
 &= \int_0^{\frac{\pi}{2}} e^{-\sin x} dx + \int_0^{\frac{\pi}{2}} e^{-\sin x} dx \text{ from (iii)} \\
 &= 2 \int_0^{\frac{\pi}{2}} e^{-\sin x} dx \\
 &< 2 \int_0^{\frac{\pi}{2}} e^{-\frac{2x}{\pi}} dx \text{ from (ii)} \\
 &= -\frac{2\pi}{2} \int_0^{\frac{\pi}{2}} -\frac{2}{\pi} e^{-\frac{2x}{\pi}} dx \\
 &= -\pi \left[ e^{-\frac{2x}{\pi}} \right]_0^{\frac{\pi}{2}} \\
 &= -\pi(e^{-1} - e^0) \\
 &= -\pi \left( \frac{1}{e} - 1 \right) \\
 &= -\frac{\pi}{e}(1 - e) \\
 &= \frac{\pi}{e}(e - 1)
 \end{aligned}$$

Hence

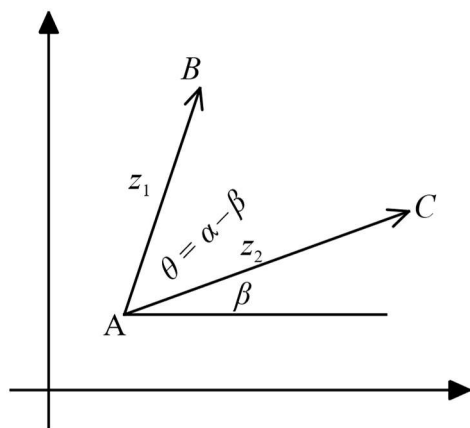
$$\int_0^{\pi} e^{-\sin x} dx < \frac{\pi}{e}(e-1) \text{ as required}$$

| Marks | Criteria                                                         |
|-------|------------------------------------------------------------------|
| 2     | Correct solution with all appropriate links clearly communicated |
| 1     | Progress towards solution                                        |

- (b) In the Argand diagram, the points A, B and C represent the triangle ABC.  
 The vectors  $\overrightarrow{AB}$  and  $\overrightarrow{AC}$  are represented by the complex numbers  $z_1$  and  $z_2$  respectively with  $\text{Arg}(z_1) = \alpha$   $\text{Arg}(z_2) = \beta$  and  $\angle BAC = \theta$ .  
 By expressing  $z_1$  and  $z_2$  in Mod-Arg form, or otherwise, show that the area of triangle ABC is given by  $\text{Area} = \left| \frac{1}{2} \text{Im}(z_1 \bar{z}_2) \right|$

3

Sample solution



$$z_1 = |z_1| \text{cis} \alpha$$

$$z_2 = |z_2| \text{cis} \beta$$

$$\begin{aligned} z_1 \bar{z}_2 &= |z_1| \text{cis} \alpha |z_2| \text{cis}(-\beta) \\ &= |z_1| |z_2| \text{cis}(\alpha - \beta) \quad (1) \end{aligned}$$

$$\therefore \text{Im}(z_1 \bar{z}_2) = |z_1| |z_2| \sin(\alpha - \beta)$$

$$\text{Area} = \frac{1}{2} |z_1| |z_2| |\sin(\alpha - \beta)| \text{ from area of a non right triangle and}$$

that the angle between the two sides of a triangle will always be acute or obtuse

$$= \frac{1}{2} |z_1| |z_2| |\sin(\alpha - \beta)| \text{ since sine of an angle in the first or second quads will be pos}$$

$$= \frac{1}{2} |z_1| |z_2| \sin(\alpha - \beta) |$$

$$= \left| \frac{1}{2} \text{Im}(z_1 \bar{z}_2) \right| \text{ as required.}$$

| Marks | Criteria                                                                                                                            |
|-------|-------------------------------------------------------------------------------------------------------------------------------------|
| 3     | Correct solution                                                                                                                    |
| 2     | Finds an expression for area of the triangle in terms of $\alpha$ and $\beta$ or finds an expression for $\text{Im}(z_1 \bar{z}_2)$ |
| 1     | Finds a simplified expression for $z_1 \bar{z}_2$ in terms of $\alpha$ and $\beta$                                                  |

(c) By establishing a reduction formula for  $I_n = \int (\sin^{-1}x)^n dx$ , show that

4

$$\int_0^1 (\sin^{-1}x)^3 dx = \frac{\pi^3}{8} - 3\pi + 6$$

Sample solution

$$I_n = \int (\sin^{-1}x)^n dx$$

$$= x(\sin^{-1}x)^n - n \int \frac{x}{\sqrt{1-x^2}} (\sin^{-1}x)^{n-1} dx$$

$$= x(\sin^{-1}x)^n - n \left[ (\sin^{-1}x)^{n-1} \sqrt{1-x^2} - (n-1) \int \frac{\sqrt{1-x^2}}{\sqrt{1-x^2}} (\sin^{-1}x)^{n-2} dx \right] + C$$

$$= x(\sin^{-1}x)^n - n\sqrt{1-x^2}(\sin^{-1}x)^{n-1} + n(n-1)I_{n-2} + C$$

$$I_1 = \int_0^1 \sin^{-1}x dx$$

$$= \left[ x \sin^{-1}x \right]_0^1 + \frac{1}{2} \int_0^1 -\frac{2x}{\sqrt{1-x^2}} dx$$

$$= \frac{\pi}{2} + \left[ \sqrt{1-x^2} \right]_0^1$$

$$= \frac{\pi}{2} + (0-1)$$

$$= \frac{\pi}{2} - 1$$

$$I_3 = \left[ x(\sin^{-1}x)^3 - 3\sqrt{1-x^2}(\sin^{-1}x)^2 \right]_0^1 - 3 \times 2I_1$$

$$= \left( 1 \times \left( \frac{\pi}{2} \right)^3 - 3(0)(0) \right) - (0 \times 0 - 3 \times 1 \times 0) - 6 \left( \frac{\pi}{2} - 1 \right)$$

$$= \frac{\pi^3}{8} - 3\pi + 6$$

$$u = (\sin^{-1}x)^n$$

$$v = x$$

$$u' = n(\sin^{-1}x)^{n-1} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$v' = 1$$

$$u = (\sin^{-1}x)^{n-1}$$

$$v = (1-x^2)^{\frac{1}{2}}$$

$$u' = (n-1)(\sin^{-1}x)^{n-2} \cdot \frac{1}{\sqrt{1-x^2}}$$

$$v' = x(1-x^2)^{-\frac{1}{2}}$$

| Marks | Criteria                                                                                                                                                           |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4     | Correct solution                                                                                                                                                   |
| 3     | Establishes formula for $I_n$ and correctly evaluates $I_1$ or,<br>Makes an error with formula for $I_1$ but goes on to follow correct process for finding $I_3$ . |
| 2     | Establishes formula for $I_n$ or correctly evaluates $I_1$                                                                                                         |
| 1     | Successfully completes one application of IBP somewhere in the problem.                                                                                            |

End of Exam